SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain the construction & Principal of an alternator. (CO7)

Q.37 Discuss about starting methods of 3 phase synchronous motors. (CO5)

Q.38 Write short notes on. (CO8)

a) Advantages of 3f system over 1f system

b) Speed control of DC motors.

(20)

No. of Printed Pages : 4 202442/122442/062443

Roll No. ....

4th Semester/ Mechatronics
Subject: DC & AC Machines

Time : 3 Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 What is the frequency of DC? (CO2)

a) 60 hz b) 0 hz

c) 25 hz d) 50 hz

Q.2 The value of slip at synchronous speed is (CO4)

a) 0 b) 1

c) 2 d) 0.5

Q.3 The stator frame of induction motor is made of (CO4)

a) Cast iron b) Silicon steel

c) Nickel d) Silver

Q.4 The Universal motor can work on (CO6)

a) AC b) DC

c) AC & DC d) None

Q.5 Watt is the unit of (CO1)

a) Force b) Flow

c) Power d) Energy

Q.6 The rate of doing work is called (CO1)

a) Energy b) Current

c) Power d) Voltage

Q.7 The unit of efficiency is (CO4)

a) Watt b) Ohm

c) Volt d) None

Q.8 Which motor runs at constant speed (CO5)

a) Synchronous b) Induction

c) Stepper d) None of above

Q.9 Which motor runs in steps (CO8)

a) Synchronous b) Induction

c) Stepper d) None of above

Q.10 The commutator in DC motors are made of _____ (CO3)

a) Copper b) Carbon

c) Gold d) Wood

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

Q.11 Define line voltage. (CO1)

Q.12 Define Phase current. (CO1)

Q.13 Define power factor. (CO1)

Q.14 The maximum value of power factor is (CO1)

Q.15 Define Back EMF. (CO3)

Q.16 Define Slip. (CO4)

Q.17 Define rotor. (CO2)

Q.18 Define Torque. (CO4)

Q.19 Brushes are made of ____. (CO3)

Q.20 Define motor. (CO3)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the importance of power factor. (CO1)

Q.22 Discuss relationship between phase current & line current in star and delta connection with diagram. (CO1)

Q.23 Write the types of DC generators. (CO2)

Q.24 Discuss the EMF equation of DC generator. (CO2)

Q.25 Explain any one method of speed control of 3 phase induction motor. (CO4)

Q.26 Write applications of 1 phase induction motor. (CO6)

Q.27 Explain armature reaction in DC generator. (CO2)

Q.28 Discuss the Torque-Slip curve. (CO4)

Q.29 Explain the working of 3 phase induction motor. (CO4)

Q.30 Write down applications of universal motor. (CO6)

Q.31 Why we need a starter in induction motors. (CO4)

Q.32 Draw the diagram of three point starter. (CO3)

Q.33 Explain working of stepper motor. (CO7)

Q.34 Explain the working of single phase induction motor. (CO6)

Q.35 Discuss applications of Servo motor? (CO8)